

Lab 8: Cellular Genetics — Mitosis and Meiosis

BIOL-1

Name: _____ Date: _____

Overview

In this lab, you will compare the two types of cell division: mitosis (for growth and repair) and meiosis (for making sex cells). You will identify the phases, track chromosome movement, and understand how meiosis creates genetic diversity.

Materials

- This worksheet
- Colored pencils (optional, for drawing)

Part 1: The Cell Cycle

1. Label each stage of interphase and what happens during it:

- G₁: _____
- S: _____
- G₂: _____

Part 2: Mitosis (PMAT)

2. Fill in what happens during each phase of mitosis:

Phase	What Happens
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Prophase

Metaphase

Anaphase

Telophase

3. What separates during anaphase of mitosis? ____
4. How many daughter cells does mitosis produce? ____
5. Are the daughter cells identical to the parent cell? ____
6. Are the daughter cells diploid or haploid? ____
7. What is the difference between a cleavage furrow (animal cells) and a cell plate (plant cells)?

Part 3: Meiosis

8. How many times does the cell divide during meiosis? ____
9. How many daughter cells does meiosis produce? ____
10. Are the daughter cells diploid or haploid? ____
11. Are the daughter cells genetically identical to each other? ____
12. What separates during Anaphase I? _____
13. What separates during Anaphase II? _____
14. What is crossing over? When does it happen?

15. What is independent assortment?

Part 4: Comparison

16. Fill in the comparison table:

	Mitosis	Meiosis
Number of divisions		
Number of daughter cells		
Diploid or haploid?		
Identical to parent?		
Purpose		

Part 5: Genetic Variation

17. Name the three sources of genetic variation in sexual reproduction:

1. _____

2. _____

3. _____

18. In humans ($n = 23$), how many unique gamete combinations are possible from independent assortment alone? (Use 2^n)

19. What is nondisjunction? What can it cause?

20. Why are you not identical to your siblings?
